

Name	Subject Class	Class	Candidate Number
	2BI		



ANGLO-CHINESE JUNIOR COLLEGE
H2 Biology Preliminary Examinations 2018

H2 BIOLOGY
 Practical

9744/04
 03 Aug 2018
2 hours and 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, index number, class, shift and laboratory on this Question Paper.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	/21
2	/20
3	/14
Total	55

This question paper consists of 19 printed pages.

Answer **all** questions.

- 1** You are required to investigate the effect of enzyme concentration on the rate of photosynthesis using plant extract after they have been incorporated into alginate beads.

The rate of photosynthesis can be measured by the changes in the colour of an indicator solution which is indicative of the amount of carbon dioxide in the solution. When carbon dioxide dissolves in water, carbonic acid is formed, thus reducing the pH and changing the colour of the indicator solution. The indicator also changes colour when carbon dioxide is removed from water.

At atmospheric concentration of carbon dioxide, this indicator is deep red in colour.

You are provided with:

- Sodium alginate, **A**
- Indicator solution, in a container labelled **B**
- Calcium chloride solution, in a container labelled **C**
- Section of a fresh leaf

Before starting the experiment, read through Steps 1 to 12.

Proceed as follows.

- 1** Place 2g of a fresh leaf on a white tile and carefully remove any large veins. Chop the rest of the leaf into very fine pieces and place the pieces into a boiling tube.
- 2** Using a dropper, add 7cm³ of water to the boiling tube. Carefully grind the chopped leaf in the boiling tube using a glass rod for about 5 minutes to obtain a dark green leaf extract. Pour the leaf extract into a beaker.
- 3** Using a 5cm³ syringe, transfer 3cm³ of the plant extract into a vial, taking care not to introduce air bubbles into the container. Next, transfer 3cm³ of the sodium alginate, **A**, into the vial with the plant extract, again taking care not to introduce air bubbles into the container.
- 4** Gently mix the plant extract and alginate by stirring the mixture with a glass rod. Do not mix vigorously as this may introduce air bubbles into the mixture.

- 5 Attach an empty 10cm³ syringe without its plunger vertically in the clamp stand, as shown in Fig. 1.1.

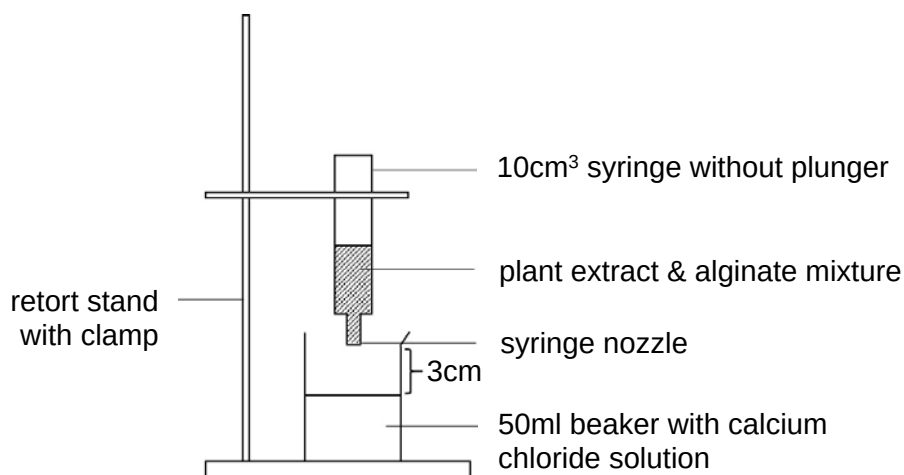


Fig. 1.1

- 6 Pour approximately 30cm³ of calcium chloride, **C** into a 50ml beaker. Place the beaker directly underneath the syringe as shown in Fig. 1.1, ensuring a distance of around 3cm between the syringe nozzle and the surface of the calcium chloride solution.
- 7 Pour the plant extract and alginate mixture into the syringe. Let all the mixture drip into the beaker of calcium chloride. The mixture will form a bead upon contact with calcium chloride. Leave the beads in the beaker of calcium chloride for 5 minutes to let them solidify.
- 8 Using a spoon, transfer the beads into an empty beaker, and rinse them using distilled water. Next, pour away the distilled water and add fresh distilled water. Leave the beads in fresh distilled water.
- 9 Using a syringe, transfer 5cm³ of indicator solution, **B** into a boiling tube. Using a pair of forceps, add 2 alginate beads containing plant extract of similar size and shape into the boiling tube. Quickly cap the boiling tube with a stopper.
- 10 Place the boiling tube into a rack and position the boiling tube 10cm from a lit lamp for 10 minutes.
- (a) (i) State and explain one modification you would make to steps 1 to 10 to increase the degree of confidence in your results.

1. Use a syringe to obtain beads of containing the same volume of plant extract

2. So that the amount of enzymes per bead is consistent/ total concentration of enzymes; OR

3. Keep the plant extract covered in foil;

4. so that reactions do not begin before the experiment begins and CO₂ is already being taken in;

5. Use cool light source/ low energy bulb/ transparent heat sink AW (A! water shield);

6. to prevent heating effect of lamp/ prevent cells from being heated (A! idea of controlling enzyme activity due to temperature); [2]

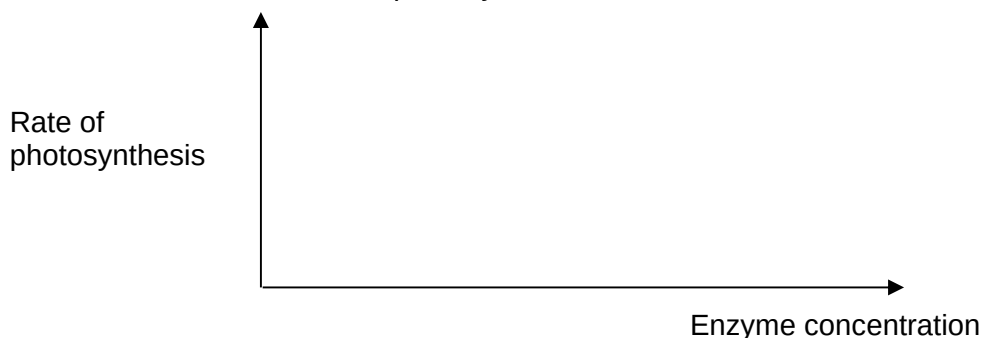
R! thermostatically controlled water bath

- (ii) Suggest a suitable control for this experiment.

Add (two) alginate beads replacing the plant extract with distilled water to indicator solution, under the same conditions as the other treatments;

[1]

- (iii) Sketch a fully-labelled graph to show the expected relationship between enzyme concentration and the rate of photosynthesis.



1. Correct axis;

2. Line starts at origin (0,0) with positive gradient, and may or may not plateau (but it should not dip);

[2]

- 11** Using a glass rod, stir the mixture in the boiling tube well to ensure that the colour of the solution is homogeneous. Next, use the colour chart to identify the pH of the mixture.

- 12** Repeat steps 9 and 10 with 6, 10, 14 and 18 alginate beads containing plant extract.

- (iv) Record your results in a suitable format in the space below.

Number of alginate beads (with plant extract)	pH of mixture
2	8.4
6	8.6
10	8.6
14	8.8
18	9.0

- 1. Correct independent variable taking first column/row, correct independent variable;**
- 2. Precision of raw data: 1dp, follow the same values on the colour chart, within the range given;**
- 3. Trend is correct; higher pH for larger number of beads;**

[3]

- (v) Explain whether your experimental results support the relationship predicted in (iii).

Yes; The pH of the indicator increases with increasing number of alginate beads; OR

No; The pH of the indicator increases with increasing number of alginate [2]

beads but no plateau is reached;

- (vi) Explain the effect of the number of alginate beads containing plant extract on the rate of photosynthesis.

1. Concentration of enzyme increases with number of beads, hence greater rate of photosynthesis;

2. Increase rate of carbon fixation in Calvin cycle/ light independent stage;

3. Increase in the number of active sites, hence increased frequency of effective collisions, hence higher rate of formation [ESC];

4. Less CO₂ in the indicator solution, hence less acidic and pH increases, colour of indicator changes;

[max 3m]

[3]

- (vii) State and explain one assumption which you made in order for a colour change to be observed at step 11.

1. CO₂ is present in the indicator solution;

2. as it is a substrate for Calvin cycle/light-independent reaction;

3. Plants cells/ chloroplasts are still intact;

4. as enzymes/ electron transport chain/ chlorophyll are required in specific order/ orientation/ position in membrane/ thylakoid lumen for specific reactions to occur;

5. Rate of respiration is not equal to the rate of photosynthesis;

6. so that there will be a change in carbon dioxide concentration which will result in a colour change;

[2]

A student carried out the same experiment as the one above except that he used a spectrophotometer to measure the colour change instead of a colour chart.

He recorded his results in Table 1.

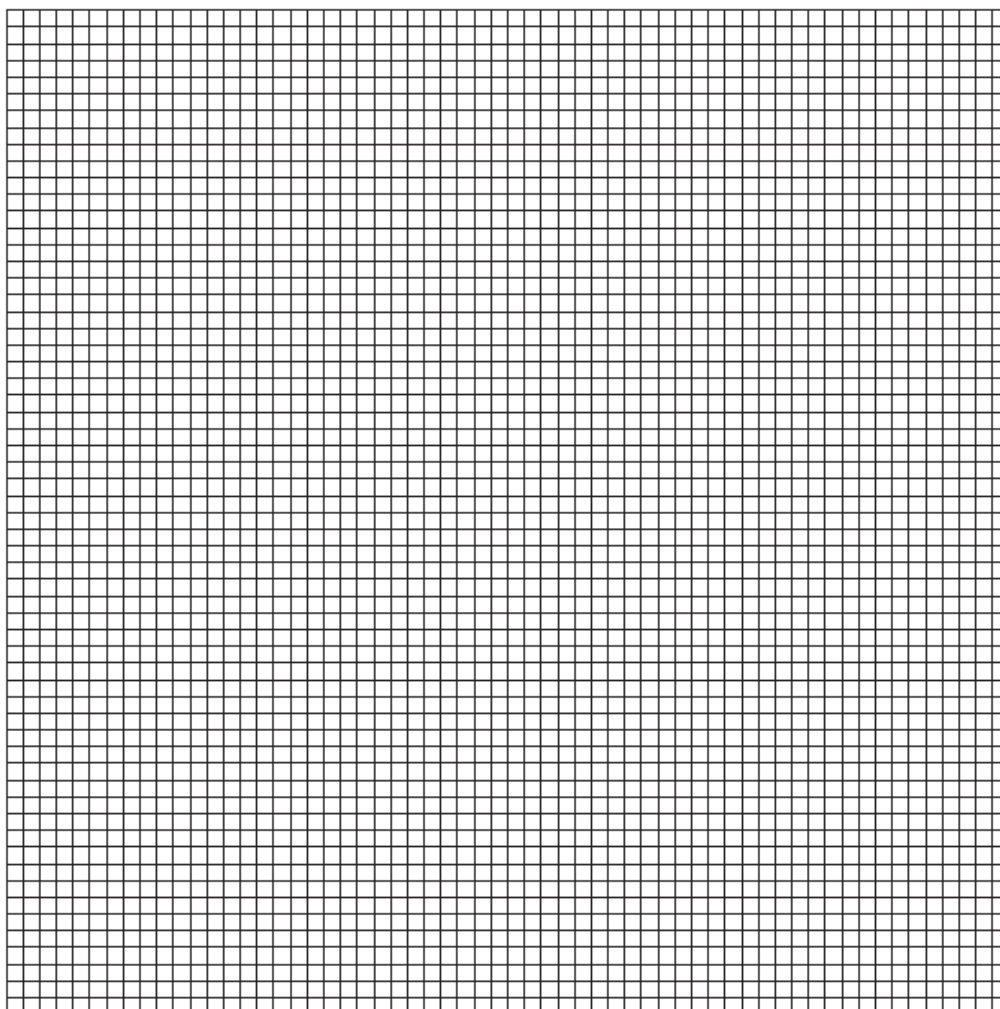
Table 1

Number of alginate beads	Absorbance / Au		
	Replicate 1	Replicate 2	Average
2	0.3655	0.3745	0.3700
6	0.6470	0.6500	0.6485
10	0.7020	0.7100	0.7060
14	0.7415	0.7485	0.7450

18	0.7620	0.7690	0.7655
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(b) (i) Complete Table 1 to show the values of the average results. [1]

(ii) Draw a graph of the student's results in Table 1 on the following grid, to show the effect of enzyme concentration on absorbance.



- Both axis correctly labelled with units, independent variable is at x-axis (x-axis: No. of beads/ enzyme concentration, y-axis: average absorbance/ Au);
- Appropriate scale: graph is at least 50% of grid and intervals are equidistant and interval markings must be shown;
- Best fit curve with no extrapolation with points plotted accurately;

[3]

(iii) Explain why the experiment is more accurate when a spectrophotometer, instead of a colour chart, is used to measure the colour change.

Visual perception of colour is subjective while spectrophotometer measures wavelength passing through solution and is more objective;

[1]

- (iv) A spectrophotometer needs to be calibrated before it is used to measure the absorbance of the solutions from the above experiment. Describe what you would use to calibrate the spectrophotometer so as to ensure a valid set of readings.

Indicator solution in a cuvette;

[1]

[Total: 21 marks]

2 Section A

In this section, you will require access to a microscope and a slide **K6**.

K6 is a stained, transverse section of a leaf blade from a plant.

Examine slide **K6** under high-power objectives (x40) of the microscope.

- (a) Use the space provided below to draw a plan diagram of part of the leaf section on slide **K6** as shown in the boxed region in Fig. 2.1. Label your plan diagram appropriately.

You should locate a suitable region of the specimen to be able to observe the distinct categories of different types of cells in the cross section.

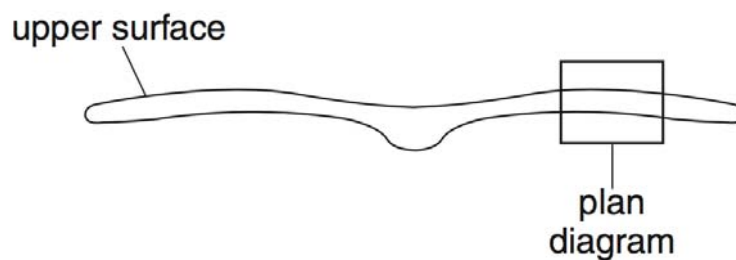
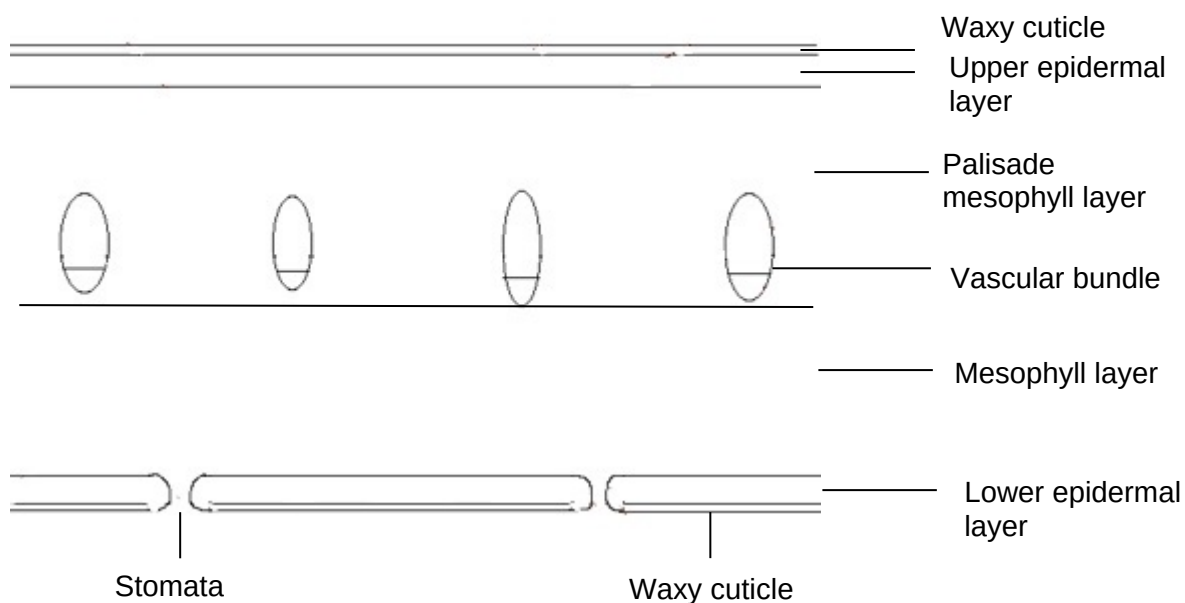


Fig. 2.1



1. **Accuracy of drawing** - Correct proportions (50-60% palisade, 40-50% spongy, 20-30% vascular bundle)
 - Show all the plan structures that can be seen in the defined part of a specimen (upper and lower epidermis, palisade and spongy mesophyll layer, vascular bundle) R! drawing of individual cells e.g. guard cells. R! too big stomata
2. **Clarity of drawing;** - Use of sharp pencil to draw clear single neat lines
3. **Scale of drawing;** - Use At least 2/3 of space provided
 - Multiple drawings are of the same scale
4. **Label of drawing;** - Clear straight lines
 - Correct plan labels (epidermis, palisade and spongy mesophyll layer, vascular bundle)

*Mark scheme taken from P10 (2017/18) and adapted

[4]

Fig. 2.2 is a photomicrograph of a stained transverse section of a leaf blade from a **different** species of plant, **K7**.

You are not expected to be familiar with this specimen.

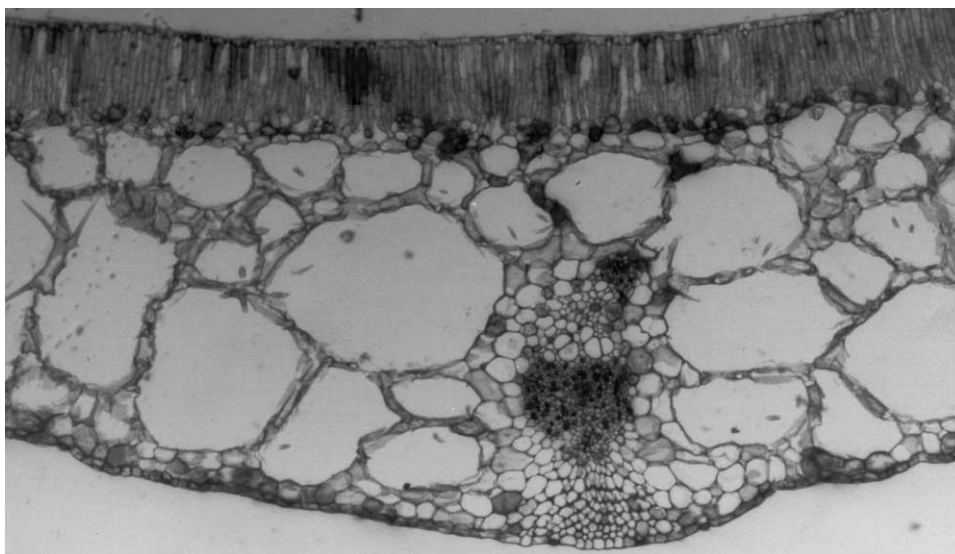


Fig. 1.2

- (b) Use a suitable table to record two observable differences between the specimen on slide **K6** and **K7**.

Comparison	K6	K7
1. Size of intercellular space	Smaller intercellular space	Larger intercellular space
2. Number of intercellular space	Fewer intercellular spaces	More intercellular spaces
3. Proportion	Spongy mesophyll layer occupies a smaller/ equal proportion than palisade mesophyll layer	Spongy mesophyll layer occupies a larger proportion than palisade mesophyll layer
4. Abundance of stomata	More stomata present	Fewer or no stomata present
5. Presence of cuticle	Thick cuticle is present	Thin/ No cuticle is present
6. Area occupied by vascular bundles	Bigger	Small

R: Regularly shaped vs irregularly shaped cells must see identity e.g. palisade and mesophyll cells

Any 2 for 2m
[2]

- (c) Based on your answer from (b), suggest the types of environment the two species of plant can be found in.

Plant species on slide K6 can be found on land while plant species on slide K7 can be found on water;

[1]

- (d) Plants with the same tissue arrangement in their leaves seen in **K7** are abundant in the type of environment they are found in.

Suggest how the abundance of these plants supports Darwin's theory of evolution.

1. The aqueous environment is the selection pressure;

2. Hence, plants with larger intercellular spaces resulting in increased buoyancy and sun exposure are selected for;

3. More of these plants survived, reproduced and pass on their alleles coding for this trait to the next generation;

4. Overtime, this results in an increase in frequencies of alleles coding for larger intercellular spaces in the gene pool of the plant population;

Max 3m

[3]

Section B

In this section, you will require access to a microscope.

You will investigate the density of stomata on the lower surface of a leaf.

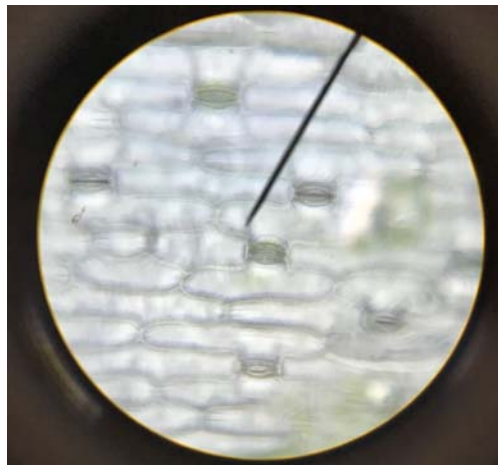
You are provided with a leaf labelled **L**.

Proceed as follows:

- 1** Cover a small area (about 1cm²) of the lower epidermis of **L** with a **thin** layer of clear nail varnish. Avoid areas with large veins. Repeat this three more times on different areas of the lower epidermis to ensure that you have spare material if necessary.
- 2** Put **L** to one side for **at least** 15 minutes to allow the nail varnish to dry.
- 3** After allowing the nail varnish to dry, carefully use the blade of a fine scalpel to lift one edge of the layer of nail varnish. You may then use forceps to gently peel the layer of nail varnish from the leaf. Ensure that the area of the layer obtained is sufficient for the field of view under the microscope. Transfer the layer to a microscope slide. Gently place a cover slip on the thin layer. No water is required.

- (a) (i)** Examine the slide under high-power objective lens (x40) and locate a stoma.

Make a detailed, labelled drawing in the space below of the stomatal pore, its guard cells and four adjacent epidermal cells.



1. Accuracy of drawing	• Draw two guard cells and four epidermal cells • Correct shape of guard cells and epidermal cells • Correct proportion of guard cells to epidermal cells (use ocular graticule to gauge proportion) • <u>Cell walls</u> shown as double lines R! if nucleus present
2. Clarity of drawing	• Use of sharp HB pencil • Clear continuous lines with no shading
3. Scale of drawing	• Use At least 2/3 of space provided
4. Label of drawing	• Correct labels (any 2 of the following – stoma/ stomatal pore, guard cell, epidermal cell);

[4]

- (ii) Calculate the **mean** number of stomata per field of view for **L** under the low-power objective lens (x10).

Show your working.

1. **Correct calculation of mean density of leaf (at least 3 readings) to nearest whole number (A! 40-150 stomata);**

Mean number of stomata per field of view for **L** =

[1]

- (iii) Using your results from (a)(ii) and given the formula

$$\text{Density of stomata (no. of stomata per mm}^2\text{)} = \frac{\text{No. of stomata}}{\text{Area of field of view,}}$$

calculate the mean density of stomata per mm² for **L**.

Show your working.

1. **Correctly calculates mean density of stomata of leaf to appropriate number of significant figures;**

Radius of field of view = 9 to 9.5 stage divisions = 0.9mm to 0.95mm

Area of field of view = $\pi (r)^2 = 2.5434$ to 2.8338mm^2

Mean density of stomata = Average no. of stomata / area of field of view

Mean density of stomata of **L** =

mm⁻²

[1]

Sun plants are plants that grow in full sunlight and are adapted to relatively high light intensities. Shade plants are not regularly exposed to full sunlight and are adapted to relatively low light intensities.

Leaves of these plants are adapted to differing light intensities. This may be shown by the density of stomata (number of stomata in a square millimeter) on the leaves.

The mean stomata density on leaves from a sun plant and a shade plant was recorded and a statistical t-test was conducted to determine whether there was a significant difference between these means.

A summary of the results is shown in Table 2.

Table 2

Mean stomatal density / mm ⁻²		Significance of difference	Total sample size
Shade leaves	Sun leaves		
290	335	$p < 0.05$	10

- (i) Comment on what these results show and suggest an explanation for any pattern.

Interpretation of statistical analysis

1. Stomatal density of sun leaves is significantly higher than shade leaves;

2. because the probability that difference in means would occur by chance is less than 5% / 0.05;

Explanation (max 2m)

3. For shade plant, light is likely to be the limiting factor;

4. while for sun plant, CO₂ concentration is likely to be the limiting factor;

5. Higher stomatal density allows higher rate of uptake of CO₂;

6. allowing sun leaves to make use of more available light for higher rate of photosynthesis;

[3]

- (ii) Describe how you can increase the statistical confidence of your conclusions.

1. Increase sample size;

R! Increase number of types of plant species tested

[1]

[Total: 20 marks]

Planning Question Marking scheme

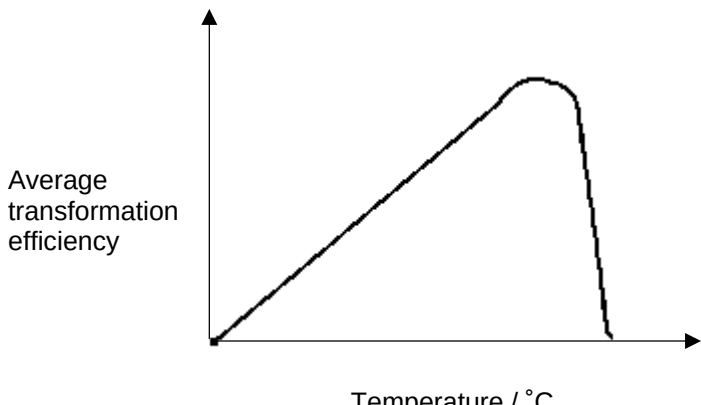
Award one mark for each of the following:

1	Independent variable	State the independent variable and use at least five different values with equal intervals, good range, units present Independent variable: temperature 30, 35, 40, 45 and 50°C;																																						
2	Independent variable	Plan suitable method to vary the independent variable Pour a mixture of hot and cold water to create a water bath of the appropriate temperatures of the 5 determined values; Or Pour water into the thermostatically controlled water bath and adjust the temperature according to the 5 determined values (preferred method);																																						
3	Dependent variable	State what is the dependent variable Dependent variable: <u>transformation efficiency as measured by the ratio of the number of transformed and viable colonies that grow on agar plate</u> (i.e. transformation efficiency);																																						
4	Dependent variable	Specifies method of measuring / monitoring the dependent variable Count the number of transformed colonies on the ampicillin plate and viable colonies growing on non-ampicillin plates;																																						
5	Controlled variables	Identifies at least two variables to control 1. Volume of <i>E. coli</i> cells / Volume of plasmid / Volume of transformation solution / Volume of LB nutrient broth used / Volume of cell suspension plated onto each petri dish (A! any volumes below 1ml); 2. Time for incubation of bacteria and plasmid in ice; 3. Constant temperature maintained in incubator; 4. Time for incubation of bacteria;																																						
6 + 7	Controlled variables	Describes how two identified variables are controlled (Must specify appropriate value): 1. Add 20µl of <i>E. coli</i> cells into microfuge tube / Add 5µl of plasmid to the tube with <i>E.coli</i> cells / Use 250µl of transformation solution / Use 250µl of LB nutrient broth / Pipette 500µl of cell suspension onto each petri dish using micropipette; 2. Incubate bacteria and plasmid in ice for about 15 – 20 minutes measured using a stopwatch; 3. Constant temperature of incubator of 37°C; 4. Incubate bacteria in the incubator overnight (e.g.12 hours) (R! anything less than 8 hours)																																						
8	Recording of results	Shows how results are to be presented in the form of a table with independent and dependent variables in appropriate columns / rows and appropriate units for headings. (ecf from method) <table><tr><th rowspan="2">Heat shock temperature / °C</th><th colspan="4">Number of colonies growing on ampicillin plates</th><th colspan="4">Number of colonies growing on non-ampicillin plates</th><th rowspan="2">Average transformation efficiency</th></tr><tr><th>Try 1</th><th>Try 2</th><th>Try 3</th><th>Mean</th><th>Try 1</th><th>Try 2</th><th>Try 3</th><th>Mean</th></tr><tr><td>30</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>35</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	Heat shock temperature / °C	Number of colonies growing on ampicillin plates				Number of colonies growing on non-ampicillin plates				Average transformation efficiency	Try 1	Try 2	Try 3	Mean	Try 1	Try 2	Try 3	Mean	30										35									
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9	Safety risks	Reference to safety hazard and precaution in the context of the experiment 1. Wear gloves and clean laboratory coat at all times to prevent bacterial infection through open wounds/ skin. 2. Dispose used gloves and disposables in an autoclave bag and autoclave before disposal to reduce contamination of equipment/environment by the bacteria; 3. Disinfect lab bench with 70% ethanol before and after experiment to prevent growth of harmful microorganisms; 4. Do not touch electric sockets of thermostatically controlled water bath with wet hands to prevent electrocution; R! Use gloves to handle hot water to prevent scalding									

Award one mark each for any FIVE of the following (max 5m):

10 + 11	Scientific reasoning	Outline of method Plasmid with ampicillin resistance gene is used to identify bacteria that have successfully taken up the plasmid via transformation. Recombinant bacteria will be able to successfully grow on ampicillin agar plates. The experiment is repeated at different heat shock temperatures to investigate its effects on transformation efficiency. Prediction of results Increase in temperature will increase kinetic energy of membrane phospholipids, increasing the rate of pore formation and DNA uptake. Increased number of recombinant bacteria colonies/ increased rate of transformation should be observed with increasing heat-shock temperature. Above optimum temperature, transformation efficiency will decrease as cell membrane is disrupted and cells die.
12	Control	Describe a suitable control Replaced plasmid solution with equivalent volume of sterile distilled water and subject to same experimental conditions (to show that survival of bacteria on LB/ampicillin plate is due to presence of plasmid)
13	Reliability	Reference to repeating at least two more times with different samples Repeat the entire experiment twice more with fresh agar plates. (can mark under table)
14	Accuracy	Reference to relevant method to increase the accuracy of results 1. Seal agar plates with parafilm to prevent contamination; 2. Sterilise measuring cylinders, micropipette tips, glass beads/ spreader, forceps before starting the experiment; 3. Ensure that there are no bacterial clumps floating in the transformation solution (to ensure cell clumps do not negatively affect the efficiency of the transformation); 4. Repeat the experiment with a narrower range of temperatures;

		5. AVP;
16	Recording of results	Shows graph with appropriate axes labels and units, line / curve with expected trend drawn  (A! Transformation efficiency and upward trend)
17	Method	Plan a method for transformation Addition of plasmid, E. coli, transformation solution/ calcium chloride and nutrient broth into microcentrifuge tube. R: Total volume that exceed 1.5ml
18	Method	Plan a method for transferring transformation mixture to agar plates Use the sterile disposable spreader to quickly spread the suspensions on the agar evenly and till most of the liquid is absorbed into the media.